

Curriculum Vitae

Personal information

- Alessandro Colombi, born in Milan, Italy, 27/12/1996.
- Living in Milan, Italy.
- e-mail: a.colombi10@campus.unimib.it

Academic positions

- Jan. 2026 - present: PostDoc Researcher in Statistics, Bocconi Institute for Data Science and Analytics (BIDSA), Bocconi University

Education

- Nov. 2021 - Feb. 2026 (expected): **Ph.D in Statistics**
University of Milano-Bicocca, Department of Economics, Management and Statistics (DEMS), Milan (Italy)
Thesis: *Bayesian learning of latent discrete structures: graphs, clusters and features allocation*
Supervisors: Raffaele Argiento - Lucia Paci
- Sept. 2018 - April 2021: **M.Sc in Mathematical Engineering**
Politecnico di Milano, Milan (Italy)
Final Mark: 110/110 cum Laude
Thesis: *Learning Block Structured Graphs in Gaussian Graphical Models*
Supervisor: Alessandra Guglielmi
Co-Supervisors: Lucia Paci, Alessia Pini
- Sept. 2015 - Sept. 2018: **B.Sc in Mathematical Engineering**
Politecnico di Milano, Milan (Italy)
Final mark: 110/110 cum Laude
Thesis: *Una soluzione esatta di un sistema preda-predatore diffusivo*
Supervisor: Maurizio Verri

Publications in refereed journals

- **Colombi A.**, Argiento R., Camerlenghi F., Paci L., Hierarchical Mixture of Finite Mixtures, *Bayesian Analysis*, 20(4), 2025, <https://doi.org/10.1214/24-BA1501>
- **Colombi A.**, Argiento R., Paci L., Pini A., Learning Block Structured Graphs in Gaussian Graphical Models, *Journal of Graphical and Computational Statistics*, 2023, <https://doi.org/10.1080/10618600.2023.2210184>.
- Codazzi L., **Colombi A.**, Gianella M., Argiento R., Paci L., Pini A., Gaussian graphical modeling for spectrometric data analysis, *Computational Statistics & Data Analysis*, 2022, <https://doi.org/10.1016/j.csda.2021.107416>.

Conference proceedings

- **Colombi, A.**, Paci, L., and Pini, A. Learning block structures in Gaussian graphical models for spectrometric data analysis. In Pollice, A. and Mariani, P., (editors), *Methodological and Applied Statistics and Demography III. SIS 2024, Italian Statistical Society Series on Advances in Statistics*, pages 444–449. Springer. ISBN 9783031644306, 2025, https://doi.org/10.1007/978-3-031-64431-3_73.
- Argiento R., **Colombi A.**, Modotti L., Montagna S., Clustering Athlete Performances in Track and Field Sports, *Book of Short Papers IES 2023*, IL VIANDANTE, pp. 23-28, ISBN 9791280333698, 2023, <https://doi.org/10.60984/978-88-94593-36-5-IES2023>.
- **Colombi A.**, Argiento R., Camerlenghi F., Paci L., Finite Mixture Model for Multiple Sample Data, *Book of Short Papers, SIS 23*, Pearson, pp. 913-917, ISBN 9788891935618AAVV, 2023.
- **Colombi A.** Block Structured Graph Priors in Gaussian Graphical Models. In: Argiento, R., Camerlenghi, F., Paganin, S. (eds), *New Frontiers in Bayesian Statistics. BAYSM 2021*. Springer Proceedings in Mathematics & Statistics, vol 405. Springer, Cham., ISBN 9783031164262, 2022, https://doi.org/10.1007/978-3-031-16427-9_6
- Codazzi L., **Colombi A.**, Gianella M., Argiento R., Paci L., Pini A., Functional Graphical Model for Spectrometric Data Analysis. In Pollice A, Salvati N., Schirripa S (Eds), *Book of short papers SIS 2020*, Pearson, pp 852 - 856, ISBN 9788891910776, 2020

Conferences and Workshops

Invited Talks

- SVEBio final workshop, January 30, 2026, Siena. How many unseen species are in multiple areas?
- GRASPA 2025, September 15 - 17, 2025, Roma. How many unseen species are in multiple areas?
- SISBAYES 2025, September 4 - 5, 2025, Padova. Learning block structures in Gaussian graphical models for spectrometric data analysis
- StaTalk 2025, June 13 - 14, 2025, Milano. Bayesian Nonparametric Inference for Unobserved Events
- 3rd ISBA Bayesian Nonparametrics Networking workshop, July 30 - August 02, 2024, Singapore. Predicting distinct and shared species in multiple areas.
- CMstatistics 2023, December 16 - December 18, 2023, Berlin. Mixture modeling via vector of normalized independent finite point processes.
- 2022 ISBA World Meeting, June 27 - July 01, 2022, Montreal. Block structured Gaussian graphical models for spectrometric functional data.

Contributed Talks/Posters

- 14th International Conference on Bayesian Nonparametrics (BNP 14), June 23 - 27, 2025, Los Angeles. Talk: How many unseen species are in multiple areas?
- BAYSM 2025, April 07 - 11, 2025, Online. Talk with Discussion: Discovering the number of species in multiple areas.
- 2024 ISBA World Meeting, July 01 - July 07, 2024, Venezia. Poster: Hierarchical Mixture of Finite Mixtures.

- The 52th Scientific Meeting of the Italian Statistical Society (SIS 2024), June 17 - 21, 2024, Bari. Talk: Learning block structures in Gaussian graphical models for spectrometric data analysis.
- BAYSM 2023, November 13 - 17, 2023, Online. Long Talk: Hierarchical Mixture of Finite Mixtures.
- Poster competition in Statistics & Probability, November 8, 2023, London. Poster: Hierarchical Mixture of Finite Mixtures.
- 2nd Bergamo Workshop in Econometrics and Statistics, September 7 - 8, 2023, Bergamo. Talk: Hierarchical Mixture of Finite Mixtures.
- Workshop sparse inference on complex networks, June 26 - 27, 2023, Lugano. Talk: Learning block structured graphs.
- Statistical Learning, Sustainability and Impact Evaluation (SIS 2023), June 21 - 23, 2023, Ancona. Talk and Poster: Finite mixture model for multiple sample data.
- Workshop Statistical methods for precise medicine, February 24, 2023, Torino. Talk: Block structured Gaussian graphical models for spectrometric functional data.
- 1st Bergamo Workshop in Econometrics and Statistics, September 15 - 16, 2022, Bergamo. Talk: Finite Mixture Model for Dependent Data.
- BAYSM 2021, September 1 - 3, 2021, Online. Talk: Learning Block Graphs Structures in Gaussian Graphical Models.
- 5th International Workshop on Functional and Operatorial Statistics, June 23 - 25, 2021, Online. Talk: Functional graphical model for spectrometric data analysis.
- 2020 ISBA World Meeting, June 28 - July 02, 2021, Online. Talk and Poster: Learning Block Graphs Structures in Gaussian Graphical Models.

Awards

- Best talk in Applications and Computation, BAYSM, 2021.
- Travel grant to attend the ISBA 2022 World Meeting in Montreal, Canada.
- Winner for *Best objective prediction* at Stats Under the Stars (SUS) hackathon, June 20 - 21, Ancona.
- Travel grant to attend the ISBA BNP Networking Workshop 2024 in Singapore.
- Travel grant to attend the 14th International Conference on Bayesian Nonparametrics (BNP 14) in Los Angeles, USA.

Research project grants

- Local unit team member, Department of Economics, Management and Statistics, University of Milano-Bicocca, PRIN 2022 PNRR project *Measuring Biodiversity via Bayesian Non-parametrics: Estimation, Clustering and Uncertainty Quantification* funded by MUR. Local coordinator Prof. Federico Camerlenghi. National coordinator Prof. Igor Prünster.

Software

- Author of GitHub repository "HSSM"
<https://github.com/alessandrocolombi/HSSM>.
- Author and Maintainer of R package "HMFm - R package for Hierarchical Mixture of Finite Mixtures",
<https://github.com/alessandrocolombi/HMFm>.
- Author and Maintainer of R package "BGSL - Bayesian Structural Learning of Block Graphs",
<https://github.com/alessandrocolombi/BGSL>.
- Co-Author and Maintainer of R package "FGM - Functional Graphical Models",
<https://github.com/TeoGiane/FGM>.

Teaching

Università Cattolica del Sacro Cuore, Milan, Italy :

Master's degree in Data Analytics for Business (LM-82)

- *Applied Linear Models* (teaching assistant, AY 2021-2022, 15 hrs; AY 2022-2023, 15 hrs; AY 2023-2024, 15 hrs; AY 2024-2025, 15 hrs)
- *Bayesian modelling* (teaching assistant, AY 2024-2025, 14 hrs)

Politecnico di Milano, Milan, Italy :

Master's degree in Mathematical Engineering (LM-44)

- *Bayesian Statistics* (tutor, AY 2021-2022, 13 hrs; AY 2022-2023 10 hrs; AY 2023-2024 10 hrs)

Referee for:

- Computational Statistics & Data Science (2024; 2025)
- Statistics and Computing (2024)

Experiences

- Lake Como school in advanced studies,
Robust Statistics: Foundations and Recent Developments, May 2-6, 2022.